

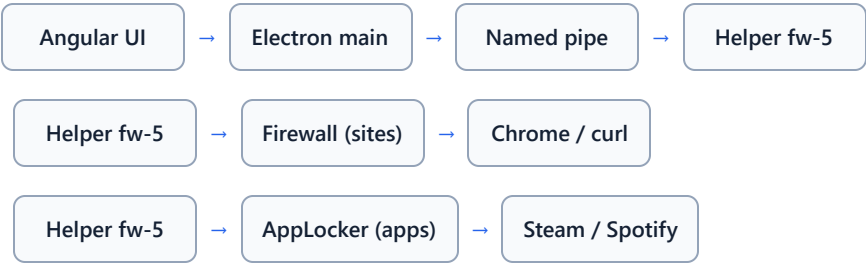
Work-session blocking (helper fw-5)

Unelevated Electron sends signed policy and session on/off over a named pipe. The LocalSystem helper enforces only while tracking is on: Firewall for sites, AppLocker Exe deny-list (Windows / Program Files / * allows plus listed denies) and Appx allow-all signed packaged apps (Settings / This PC Properties stay usable).

Two engines, one clock
Sites lose outbound connectivity (any browser or curl). Listed apps cannot CreateProcess, and already-running matches are closed once. Stop Tracking deletes both rule sets. AppLocker is never applied as deny-only — that fw-3 pattern locked Task Manager. fw-4 Exe-only (no Appx) locked Settings / This PC.



How layers communicate



What each layer owns
Angular. Banner, Start/Stop Tracking. Never opens the pipe.
Electron main. Work-session gate, Ed25519 policy, pipe client only.
Helper. Verify policy, DNS, INetFw, AppLocker XML, one-shot close.
OS. Firewall drops packets. AppLocker refuses launch.

IPC allowlist
No RunCommand. A compromised UI must not get a SYSTEM shell.
ApplyPolicy SetSession GetStatus GetRecentBlocks
ReportTamper

Tech stack

Layer	Stack
UI	Angular 21 + TypeScript
Desktop	Electron 42 (main + preload)
Helper	Rust, windows 0.58, windows-service, tokio
Policy	JSON + Ed25519 (dalek / Node crypto)
IPC	Named pipe, length-prefixed JSON, SDDL
Sites	DNS → INetFw outbound RemoteAddresses
Apps	AppLocker Exe+Appx deny-list + one-shot close
Ship	electron-builder NSIS per-machine

Session gate

Enforcement requires helper LocalSystem (or admin), a signed policy in block mode, and an active work session.

session off → clear rules

policy missing → idle

session + policy → fw + AppLocker

Technical issues

Issue	Why	What we do
AppLocker missing (Home)	Application Identity / AppLocker not on the SKU	last_error; already-running apps still closed once; new launch may succeed
Leftover rules after uninstall	INetFw and AppLocker persist after the service is deleted	--clear-blocks / --uninstall wipe VMG Sentinel* rules
CDN / DoH miss	Chrome may use an IP the helper did not resolve	Extra host prefixes; still IP-based, not SNI
Wrong helper on the pipe	Old WFP binary or user-level spawn	Refuse attach unless helper_build is fw-5
This PC / Settings blocked (fw-4)	Exe Enabled without Appx allow-all	fw-5 enables Appx allow all signed packaged apps
Tracking on, nothing blocked	SetSession without ApplyPolicy	Apply local deny list if policy_loaded is false
Rename / portable exe	Default list is still *\steam.exe; Citadel can add hash/publisher	Helper emits FileHashRule / FilePublisherRule when policy includes them
Deny-only AppLocker (fw-3)	Enabled Exe collection with no Allow * default-denies the PC	fw-5 writes Windows / Program Files / * allows first; apply refuses without them and without Appx allow
IT already has AppLocker	Merge/remove must not wipe their rules	Only add/remove rules named VMG Sentinel *; clear sets Exe and Appx NotConfigured
Pro launch deny after install	AppLocker may not enforce until reboot	Reboot once; Home still fail-open
SmartScreen on Setup.exe	Unsigned installer	More info → Run anyway until Authenticode

Out of scope

No TLS MITM, no hosts file, no IFEO, no WDAC XML deploy, no taskkill loop, no macOS NEFilter yet. All-day “never run this exe” still belongs in Intune/WDAC, not this session switch.

PDF counterpart of the canvas “Work-session blocking (helper fw-5)”. Aligns with HELPER_BUILD fw-5. 25 September 2026. Regenerate the PDF from this HTML.